

QENA SOUTH VALLEY UNIVERSITY, Egypt

WMO: 624030

Lat: 26.2003N

Lon: 32.7467E

Elev: 254

StdP: 14.56

Time Zone: 2.00 (E02)

Period: 04-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.8	46.8	12.6	10.5	81.1	16.4	12.7	80.4	17.1	60.9	15.1	63.2	2.5	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	108.8	66.2	106.8	66.2	105.0	66.2	72.5	98.9	70.8	97.7	69.5	96.8	8.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	86.1	87.4	60.3	78.9	86.3	58.4	73.5	85.7	36.2	98.7	34.8	97.2	33.7	96.9	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	17.4	15.5	13.8	DB	41.2	111.6	2.4	2.4	39.5	113.3	38.1	114.7	36.7	116.1	35.0	117.9
(5)				WB	34.2	77.0	2.1	2.4	32.7	78.8	31.4	80.2	30.2	81.6	28.7	83.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.5	59.4	64.6	71.1	80.4	86.7	91.7	92.3	91.8	88.2	82.3	71.1	62.2
	DBStd	12.68	4.81	5.87	5.87	5.98	5.12	3.61	2.69	2.57	3.44	5.13	5.56	4.79
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	394	190	72	9	0	0	0	0	0	0	0	7	116
	CDD50	10416	290	408	655	911	1137	1252	1310	1297	1145	1001	632	378
	CDD65	5335	15	59	200	461	672	802	845	832	695	536	189	29
	CDH74	78013	150	645	2138	5938	10153	13266	13924	13419	10098	6434	1607	241
	CDH80	48761	15	202	853	3315	6439	9085	9512	9028	6235	3476	550	51
Wind	WSAvg	5.8	4.2	5.0	5.6	6.3	6.5	7.1	7.1	7.2	6.9	5.2	4.5	4.2
Precipitation	PrecAvg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	0.5	0.1	0.1	0.1	0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.0	91.0	98.8	104.8	110.3	112.7	109.7	108.9	107.0	104.2	93.9	85.6
		MCWB	60.4	63.0	63.3	64.1	65.6	66.1	67.2	67.8	66.4	65.7	63.0	60.4
	2%	DB	78.6	86.1	93.0	101.4	106.2	109.5	107.3	106.4	104.4	100.3	89.8	80.3
		MCWB	58.2	60.0	60.4	62.6	64.6	65.7	67.1	67.6	66.8	65.6	63.5	59.6
	5%	DB	74.3	81.6	88.2	98.4	103.4	107.1	105.3	104.7	101.9	97.2	86.2	76.0
		MCWB	55.3	57.7	58.2	61.5	63.7	65.3	67.1	67.4	66.4	65.7	62.4	57.2
	10%	DB	70.6	77.4	84.3	95.0	100.5	104.4	103.5	103.0	99.7	94.4	82.6	72.9
		MCWB	53.6	55.5	57.7	60.5	62.4	64.7	66.7	67.2	66.5	65.3	60.7	55.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.6	64.3	67.9	68.1	67.5	70.4	74.9	75.1	73.8	71.4	68.1	64.0
		MCDB	77.3	88.1	83.6	91.8	100.2	101.7	101.5	101.2	96.1	94.0	86.6	78.7
	2%	WB	59.3	61.1	63.3	64.7	66.0	68.3	72.0	72.5	71.0	69.0	65.2	60.7
		MCDB	74.7	81.5	87.7	95.0	102.0	100.2	99.9	99.3	96.3	92.9	84.6	75.5
	5%	WB	56.5	58.5	60.5	62.9	64.7	67.1	70.2	70.7	69.6	67.7	63.4	58.5
		MCDB	71.3	79.4	86.4	94.9	99.9	100.0	98.3	98.2	95.4	91.1	82.4	73.8
	10%	WB	54.5	56.3	58.2	61.3	63.5	66.0	68.8	69.3	68.3	66.4	61.7	56.7
		MCDB	68.7	75.6	82.4	92.7	97.5	99.7	97.5	97.4	94.0	90.4	80.6	71.2
Mean Daily Temperature Range	5% DB	MDBR	22.0	23.3	24.2	25.2	25.0	24.9	23.0	23.5	23.6	22.9	22.4	21.9
		MCDBR	25.2	27.2	27.7	27.6	28.1	27.3	24.4	24.7	25.3	24.7	24.6	24.5
		MCWBR	13.8	12.9	12.1	10.8	9.7	8.7	7.2	7.7	7.9	8.7	10.9	12.4
	5% WB	MCDBR	23.3	25.8	25.0	25.5	25.8	25.0	22.6	23.2	23.3	22.8	23.0	23.2
	MCWBR	13.6	12.8	12.0	11.0	9.3	8.4	8.3	9.1	8.5	8.6	10.6	12.0	
Clear-Sky Solar Irradiance	taub	0.368	0.400	0.443	0.527	0.541	0.461	0.469	0.470	0.456	0.458	0.384	0.365	
	taud	2.194	2.077	1.948	1.762	1.748	1.967	1.977	1.980	1.997	1.973	2.171	2.212	
	Ebn at Noon	276	275	270	250	246	265	262	262	262	253	268	272	
	Edh at Noon	40	49	59	73	74	59	58	57	55	53	41	38	
All-Sky Solar Radiation	RadAvg	1410	1708	2067	2328	2472	2592	2510	2362	2131	1787	1492	1321	
	RadStd	51	58	62	46	64	25	32	35	48	36	29	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.03	N/A	N/A	+1.03	+1.22	+1.15	N/A	-63	+418	+369

Nomenclature: See separate page